K. Sandhya
192472207

CO2 - AT3

class Test

22/25

① Concept of Image enhancement in the spatial domain.

Image enhancement in the spatial domain improve the visual quality of an image by directly modifying the pixel intensity values. It makes images clearer and remove unwanted effects.

a) Gray-level Transformations

These techniques modify pixel brightness and contrast.

Examples : Image Negative, Log Transformation, contrast stretching.

Use : Enhances dark images. improves visibility and highlights important details.

b) Histogram processing

A histogram shows the distribution of pixel intensities.

Histogram Equalization : Improves overall contrast by redistributing intensity values.

Histogram Matching : Adjust the histogram to match a desired distribution.

Histogram stretching: Expands the intensity range to im[...]
contrast.

c) Spatial filtering

use a filter mask to process neighboring pixels.

Smoothing filters: Reduce noise and blur the image.

Sharpening filters: Enhance edges and fine details.

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(2)

Image smoothing	Image sharpening
* Image Smoothing is Remove noise and small variation	* Image sharpening enhances edges and fine details.
* use low-pass filters such as Mean and Gaussian filter.	* uses High-pass filters such as laplacian and sobel operator.
* Averages neighboring pixels	* Highlights intensity changes between neighboring pixels.
* produce a Smoother image but reduces edge sharpening.	* produces clearer object boundaries.
Application: Medical image, Satellite image, preprocessing	Application: object detection, OCR defect detection,

working principle of Spatial smoothing filter

* Each pixel is replaced by average or weighted average of its neighboring pixels.
* this reduces sudden intensity Variations and remove noise.

Techniques:

* Mean filter
* Median filter
* Gaussian filter

working principle of sharpening filter

* A sharpening kernel moves across the image. It detects rapid intensity changes.
* ~~Edge pixels~~ are slowly changing region remain less affected.

Techniques:

* laplacian filter
* Prewitt filter
* Sobel filter
* High-Boost filter

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3 Image enhancement process in the frequency domain.

It is performed by first converting the image from spatial domain to frequency domain using fourier Transform (FFT). After applying the required filter, the image is converted back using Inverse fourier Transform (IFFT)

Low pass filter (LPF)

* Allow low frequency components to pass and blocks high frequency components

* Removes noise and smooth the image and produce
blur effect.

High pass filter (HPF):

* Allows high- frequency components to pass and blocks low-frequency components.
* Makes the image sharper.

Comparison :

LPF :	HPF:
Noise reduction, smooth image, removes high frequency noise	Edge enhancement, share image, highligths fine details.

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(4) Concept of Image Segmentation

Image segmentation is the process of dividing an image into meaningful regions or objects so that each region can be analyzed separately.

1. point detection

Detects isolated pixels with significantly different intensity.

Advantage : Simple and fast

Limitation : Sensitive to noise

Application : Detecting defective pixels.

2. Line detection

Advantage : Identifies linear structures.

Limitation : May miss curved objects.

Application : Road and lane detection.

3. Edge Detection

Detects object boundaries using operators like sobel, prewitt, and canny.

Advantage : Accurate boundary detection

Limitation : Noise may create false edges.

Application : Medical imaging and object recognition.

4. Thresholding

separates foreground and background using Intensity values

Advantage : simple and computationally efficient.

Limitation : performs poorly under uneven lighting.

Application : Document scanning and binary Image creation.

5. Region-Based segmentation

Groups neighboring pixels with similar properties.

Advantage : produces complete object

Limitation : computationally expensive

Application : Medical image analysis and satellite image segmentation.

(5) Importance of Edge Detection, Edge linking and boundary detection.

1. Edge Detection

* Identifies sudden changes in image intensity.
* Detect object boundaries.
* Common operators: Sobel, Prewitt, Canny
* Example: Detecting cracks in roads

2. Edge linking

* Connects broken or discontinuos edges to form continuous object boundaries.
* Improves object recognition accuracy.
* Example: joining broken edges of a vehicle in traffic image

3. Boundary Detection.

* Extract the complete outline of an object after edge detection and linking.
* Example: Detecting tumor boundaries in MRI scans.

<u>Flowchart</u>:

Input image
↓
Edge Detection
↓
Edge Linking
↓
Boundary Detection
↓
Object Recognition / Defect Detection